

Golf Cart Diagnostic Technician — Written Final Exam

Version: 1.1 · 50 questions · Passing score **70% (35/50)**

Time allowed: 40 minutes

Materials allowed: None (closed book) unless instructor authorizes formula card

Student name: ____ **Date:** _

Instructor: ____ **Score:** _ / 50 = ____%

Multiple Choice (1-40) — Circle the best answer

1. Before probing a battery pack you believe is disconnected, you should:
 - A) Trust the key-off position
 - B) Verify zero energy with a meter after isolation
 - C) Remove the motor first
 - D) Wet the terminals
2. Ohm's Law states:
 - A) $V = I \times R$
 - B) $P = R / V$
 - C) $I = R \times P$
 - D) $R = I / V$ always only
3. Six 8V batteries in series produce a nominal:
 - A) 8V
 - B) 24V
 - C) 48V
 - D) 96V
4. Hydrogen gas hazard is most associated with:
 - A) Tire inflation only
 - B) Battery charging without ventilation
 - C) Regenerative braking heat only
 - D) Steering play
5. Continuity beep alone proves a 2/0 battery cable:
 - A) Can carry full traction current with low drop
 - B) Is not open — but not that it is adequate under load
 - C) Has correct SG
 - D) Has correct brush length

6. Specific gravity testing applies primarily to:
 - A) Lithium pouch cells
 - B) Flooded lead-acid cells
 - C) Curtis programmers
 - D) Gasoline octane
7. A battery that passes rested OCV but fails load test is often:
 - A) Perfect
 - B) Unable to deliver current (e.g., sulfated/high internal resistance)
 - C) Always a bad controller
 - D) Always low tire pressure
8. Equalization charging is generally for:
 - A) Lithium weekly
 - B) Flooded lead-acid when procedure indicates
 - C) All chemistries identically
 - D) Transaxles
9. Using a lead-acid charger on a lithium pack is:
 - A) Recommended if voltages match nominally
 - B) Unsafe / incorrect — chemistry profiles differ
 - C) Required for balancing steel
 - D) Only for gas carts
10. BMS functions commonly include:
 - A) Cell balancing and over/under voltage protection
 - B) Tire rotation
 - C) Carburetor idle set
 - D) Kingpin torque only
11. Series-wound and SepEx motors differ mainly in:
 - A) Paint color
 - B) Field excitation arrangement/control
 - C) Fuel type
 - D) Number of tires
12. Brushes are measured because:
 - A) Length relates to remaining life vs OEM limits
 - B) They set tire pressure
 - C) They replace fuses
 - D) They store fault codes
13. Insulation-to-ground checks on a motor should be done:
 - A) While meggering through a live controller carelessly
 - B) With motor properly isolated from sensitive electronics
 - C) Only underwater
 - D) Using a hydrometer

14. PWM in a controller primarily:
- A) Waters batteries
 - B) Modulates power to control motor speed/torque
 - C) Measures SG
 - D) Aligns steering
15. High Pedal Disable (HPD) is meant to:
- A) Raise top speed
 - B) Inhibit drive if throttle is applied in an improper power-up sequence
 - C) Bypass solenoid contacts forever
 - D) Equalize flooded cells
16. A solenoid that clicks but does not pass current well may have:
- A) Perfect contacts
 - B) Failed/high-resistance contacts
 - C) Infinite coil resistance always
 - D) A good brush length only
17. Curtis LED flash codes should be interpreted with:
- A) Random guessing
 - B) Model-correct documentation / Curtis resources
 - C) Tire charts
 - D) Octane ratings
18. Potentiometer throttles are commonly checked by:
- A) Compression testing
 - B) Measuring resistance or voltage through pedal travel
 - C) SG only
 - D) Transaxle backlash only
19. The first diagram rule is:
- A) Any year diagram is fine
 - B) Match make/model/year (configuration)
 - C) Ignore OEM legends
 - D) Colors are universal across brands
20. Best check for a main cable under load:
- A) Voltage drop while current flows
 - B) Smell only
 - C) Paint thickness
 - D) Radio volume
21. An "open" in a circuit means:
- A) Unintended path to ground
 - B) Broken/incomplete path
 - C) Perfect continuity
 - D) High SG

22. Step 1 of the 7-step diagnostic process is:
- A) Order a controller
 - B) Verify the complaint
 - C) Skip to road test
 - D) Disable HPD
23. Step 7 is:
- A) Guess
 - B) Verify repair and road test as applicable
 - C) Delete documentation
 - D) Bypass fuses
24. "Diagnose before replace" means:
- A) Swap parts until something works
 - B) Prove root cause with inspection/tests before condemning parts
 - C) Never replace anything
 - D) Only replace batteries
25. On many electric "no go/slow" complaints, a high-yield starting area is:
- A) Seat upholstery brand
 - B) Batteries and high-current connections
 - C) Wheel color
 - D) Radio presets
26. Regenerative braking:
- A) Eliminates need for friction/parking brakes
 - B) Works with mechanical brakes; does not replace them
 - C) Waters cells
 - D) Is only on gas carts
27. Before working under a raised cart:
- A) Jack alone is enough
 - B) Use rated stands on proper points + chocks
 - C) Have someone bounce the cart
 - D) Remove stands for clearance
28. Tire pressure affects:
- A) Handling and effective range
 - B) Only paint shine
 - C) Controller CAT rating
 - D) Spark plug heat range only
29. A typical training-program gas golf cart engine is:
- A) Jet turbine
 - B) Single-cylinder 4-stroke (common OEM types)
 - C) Pure electric only
 - D) V12

30. Carbon monoxide risk is associated with:
- A) Running gas engines without proper ventilation/exhaust control
 - B) Measuring continuity
 - C) Reading diagrams
 - D) Checking tire date codes
31. Classic gas no-start pillars include:
- A) Spark, fuel, compression (plus air/interlocks)
 - B) Only lithium cell balance
 - C) Only PWM percent
 - D) Only SG
32. Daily PM commonly includes:
- A) Full armature rewind
 - B) Tires, obvious damage, brakes, unusual noises, basic battery cues
 - C) Controller redesign
 - D) Disabling interlocks
33. Club Car, EZ-GO, and Yamaha are referenced as:
- A) Tire brands only
 - B) Major golf cart OEMs for manuals/brand practices
 - C) Hydrometer brands
 - D) Fuel octane levels
34. Class 00 gloves in this program relate to:
- A) Electrical PPE when used as rated
 - B) Welding only
 - C) Rain fashion
 - D) Transaxle fill only
35. Parallel batteries primarily:
- A) Add voltage
 - B) Add capacity (Ah) at same voltage
 - C) Reduce Ah
 - D) Create gasoline
36. A charger “fan running” proves:
- A) Correct DC current into the pack
 - B) Only that the fan circuit may be active — still measure DC performance
 - C) HPD is bypassed
 - D) Compression is high
37. Dielectric grease:
- A) Replaces metal contact in a bad crimp
 - B) Helps exclude moisture after a proper connection is made
 - C) Is battery electrolyte
 - D) Is gear oil

38. Intermittent faults are often found at:
- A) Random planets
 - B) Connectors, chafe points, poor grounds, heat-sensitive joints
 - C) Only brand-new controllers
 - D) Only tire valve caps
39. Professional documentation should avoid:
- A) Units on measurements
 - B) Customer passwords / unnecessary PII
 - C) Fault codes
 - D) Verification notes
40. Fuses should be replaced with:
- A) Copper slug / foil
 - B) Correct type and amp rating
 - C) Any larger fuse for convenience
 - D) A bolt
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Short Answer (41-50) — Write clearly

1. Write Ohm's Law and calculate current if $V = 36V$ and $R = 12\Omega$.
 2. List the seven diagnostic steps in order.
 3. Name three PPE items required for battery pack work.
 4. Why is the correct model-year wiring diagram required?
 5. What does a high solenoid contact voltage drop under load indicate?
 6. Give two differences between flooded lead-acid and lithium ($LiFePO_4$) maintenance/charging.
 7. Describe a safe operator sequence concept to avoid HPD trips (general).
 8. List three mechanical systems inspected in Week 8.
 9. Why ventilate battery charging areas?
 10. State the program mantra that discourages parts swapping without evidence.
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End of exam. Do not write on the answer key.